

# My Spacecraft to Saturn

*Illustration of Cassini approaching Saturn.*



## LESSON TIME

1 hour

## MATERIALS CHECKLIST

*For the teacher:*

- Copy of “Summary of Spacecraft Design Problems” to read aloud

*For each student:*

- Copy of “Designing Your Spacecraft Worksheet”
- Pencil
- Paper
- Saturn Discovery Log

## LESSON NO. 6

- *Language Arts Focus*  
—Nonfiction Reading and Writing as Tools for Problem-Solving  
—Nonfiction Reading Practice: Summary
- *Science Focus* — *Thinking Like an Engineer: Problem-Solving Spacecraft Design*

### OVERVIEW

In this lesson, students get the chance to think like engineers as they are presented with problems that the NASA team faced when designing a spacecraft to travel to Saturn. Students work with partners to think of solutions to address those problems, and use these ideas to sketch their spacecraft. In the next lesson, students have the opportunity to compare their ideas to the ideas of the NASA team. This is a unique opportunity for children to gain an insider’s look at the world of engineering.

### WHY THIS WORKS

The National Science Education Standards state that children should be able to design solutions to problems. This exposure to design in the younger grades provides the groundwork for more sophisticated work in later grades. Drawing illustrations with accompanying text is a powerful way for children to organize and express their thinking. Nonfiction writers often rely on illustrations with text to explain difficult concepts, so students benefit from repeated exposure to creating and learning from illustrations with text.

## Objectives

Students will:

- Practice working with a partner to think of ideas to solve problems.
- Practice communicating solutions clearly.
- Practice communicate ideas using illustrations with text.

## Teacher Preparation

Print out a copy of teacher reference 1: “Summary of Spacecraft Design Problems” to read aloud. Make a copy for each student of student handout 1, “Designing Your Spacecraft Worksheet” (4 pages). Decide how to pair students.

